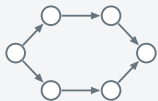


1. Rollout

 π_t 

Rollout + GAE

 A_r, A_{c_k}

$\bar{J}_k$ observed
episode cost

2. Flow inputs



Clustered states
Width $\propto$ flow mass

 $\hat{F}, A, C_k$

3. Constraints

Soft constraints

$$\langle C_k, F \rangle \leq b_k$$

$$b_k = B_k/L$$



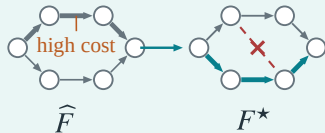
$$F_H = 0$$

Hard constraints

4. SafeOT projection

$$\min_{F \geq 0} -\langle A, F \rangle + \varepsilon \text{KL}(F \| \hat{F})$$

s.t. flow conservation + constraints



Target flow is not executed

5. Priced update

$$\beta_k \leftarrow [\beta_k + \alpha(\bar{J}_k - B_k)]_+ \rightarrow \oplus \rightarrow \lambda_k \rightarrow \tilde{A} = \frac{A_r - \sum_k \lambda_k A_{c_k}}{1 + \sum_k \lambda_k} \rightarrow \text{PPO/TRPO} \rightarrow \pi_{t+1}$$

$$\bar{\lambda}_k^* \leftarrow \text{clip} + \text{EMA} \leftarrow \lambda_k^*$$

safety prices

next rollout